

$$\begin{array}{c} R^3 \\ \Rightarrow \\ S \subset \\ R^3 \\ S \\ S: \\ \Sigma \rightarrow \\ S \\ S \\ S \\ \varphi(\overline{\Sigma}) \subset \\ R^3 \\ S \end{array}$$

Mainardi-Codazzi

$$\begin{array}{c} S \\ p \in \\ S \\ K(p) > \\ 0 \\ p \\ k_1 \\ k_2 \\ k_1 \geq \\ k_2 \\ p \\ S \\ (u,v) \\ F = \\ f = \\ 0 \\ k_1 = \\ e/Ek_2 = \\ g/G \end{array}$$

$$e_v=\frac{E_v}{2}(k_1+k_2),g_u=\frac{G_u}{2}(k_1+k_2).$$

$$-2(k_1-k_2)KEG=-2E(k_1)_{vv}+2G(k_2)_{uu}+\cdots.$$

$$\begin{array}{c} p \\ (k_1)_v = \\ 0(k_2)_u = \\ 0(k_1)_{vv} \leq \\ 0(k_2)_{uu} \geq \\ 0 \\ K > \\ 0k_1 > \\ k_2 \\ p \\ 1 \end{array}$$

Example

$$1 \\ x=C\cos v\cos u,\;y=C\cos v\sin u,\;z=\psi(v),0<u<2\pi,\;|v|<\sin^{-1}(1/C),$$

$$\begin{array}{c} \varphi(v) = \\ C\cos v\psi(v) = \\ \int \sqrt{1-C^2\sin^2v}\,dvC > \\ 1 \end{array}$$

$$\begin{array}{c} E = \\ C^2\cos^2vF = \\ 0G = \\ 1e = \\ -C\cos v\sqrt{1-C^2\sin^2v}f = \\ 0g = \\ -C\cos v/\sqrt{1-C^2\sin^2v} \end{array}$$

$$\begin{array}{c} K = \\ k_1k_2 = \\ 1 > \\ 0 \\ k_1 \neq \\ k_2 \\ k_1 = \\ 0 \\ 2 \end{array}$$

$$\begin{array}{c} S \subset \\ R^3 \\ S \\ S \\ S \\ \Sigma \\ S \\ S \\ p \\ S \\ \Sigma \\ K_S(p) \geq \\ K_\Sigma(p) > \\ 0 \\ p \\ 3 \\ ?? \end{array}$$

$$\begin{array}{l} k_2(q) \\ k_1(p) \geq \\ k_1(q) \geq \\ k_2(q) \geq \\ k_2(p) = \\ k_1(p) \\ k_1(q) = \\ k_2(q) \\ S \\ S \\ K > \\ QS \\ S \\ S \\ S \\ S = \\ S \\ S = \\ k_2 = \\ K/k_1 \\ k_1 \\ H = \\ \frac{1}{2}(k_1 + \\ k_2) \\ k_2 = \\ f(k_1) \\ S \\ K > \\ Q \\ H \\ S \\ K > \\ 0 \\ ?? \end{array}$$

$$\frac{??}{S} \quad .$$

$$\begin{array}{l} \overline{S} \subset \\ R^3 \\ p_0 \notin \\ S \\ d(q) = \\ \frac{1}{2}|q - \\ p_0|^2 q \in \\ S \\ q_0 \in \\ S \\ d(q_0) \geq \\ d(q) \\ q \in \\ S \\ q_0 \\ S \\ ?? \\ d(q) = \\ \frac{1}{2}|q - \\ p_0|^2 \\ S \\ p_0 \\ q_0 \\ S \\ p_0 \\ ?? \\ q_0 \\ q_0 \in \\ S \\ d(q_0) \geq \\ d(q) \\ q \in \\ S \\ N(q) \\ S \\ p_0 \end{array}$$

$$d(q)=\frac{1}{2}\langle q-p_0,\,q-p_0\rangle,\nabla d(q)=q-p_0.$$

$$\begin{array}{l} q_0 \\ d|_S \\ \langle q_0-p_0,\,w\rangle=0\forall\,w\in T_{q_0}(S). \end{array}$$

$$\begin{array}{l} q_0- \\ p_0 \\ T_{q_0}(S) \\ q_0- \\ p_0 = \\ \pm|q_0- \\ p_0|N(q_0) \end{array}$$

$$d(q_0)=\frac{1}{2}|q_0-p_0|^2=\frac{1}{2}R^2,$$

$$\begin{array}{l}
S\\
\mathcal{P}_S\\
\mathcal{P}_S\\
\mathcal{P}_v\in\\
T_p(S)|v|=\\
\gamma_v\\
\mathcal{P}_v\\
\gamma_v\\
d(p,\gamma_v(s))=\\
s\geq\\
0\\
s_v>\\
0\\
d(p,\gamma_v(s_v))<\\
s_v\\
T_p^1(S)\\
S_1^1\\
s_v>\\
0\\
s_v\\
\gamma_v\\
[0,s_v]\\
s_v\\
\mathcal{P}_v\\
\gamma_v\\
d(p,\cdot)\\
q_n\in\\
S\\
d(p,q_n)\rightarrow\\
\infty\\
\mathcal{P}_n\\
q_n\\
l_n=\\
d(p,q_n)\\
v_n\in\\
T_p^1(S)\\
S_1^1\\
v_{n_k}\rightarrow\\
v\gamma_{n_k}\rightarrow\\
\gamma_v^v\\
\mathcal{P}_v^v\\
s_0\\
d(p,\gamma_v(s_0))<\\
s_0\\
n_k>\\
l_{n_k}>\\
s_0\\
\gamma_{n_k}^v\\
S\subset\\
R^3\\
S\\
\alpha:[0,\infty)\rightarrow\\
S\\
|\alpha'|\\
\mathfrak{A}\\
S\\
\mathcal{P}:\\
\gamma:[0,\epsilon)\rightarrow\\
Sl(\gamma)=\\
\mathfrak{s}=\\
\mathfrak{e}+\\
\delta\\
\delta_n\rightarrow\\
0\\
p_n=\\
\gamma(\epsilon-\\
\delta_n)\\
p_n\rightarrow\\
p_\epsilon=\\
\lim_{s\rightarrow\epsilon}\gamma(s)\in\\
R^3\\
p_\epsilon\notin\\
S\\
\beta_n\\
p_n\\
\gamma\\
p_\epsilon\\
\beta_n\\
S\\
\mathcal{P}\in\\
S\\
\exp_p\\
v\in\\
T_p(S)\\
|v|=\\
\mathcal{P}_n\rightarrow\\
v\mathcal{P}_n\in\\
T_p(S)|v_n|<\\
r\\
p_n=\\
\exp_p(v_n)\in\\
S
\end{array}$$

$$w'(t)=-K(t)-\frac{1}{4(t+1)^2}, w(t)^2=\left(\int_t^\infty K+\frac{1}{4(t+1)}\right)^2\geq \frac{1}{4(t+1)^2},$$

$$\begin{array}{l} w'(t)+\\ w(t)^2\leq \\ -K(t)\\ v(t)\\ v'(t)=w(t)v(t), v''(t)=(w'(t)+w(t)^2)v(t)=-L(t)v(t). \end{array}$$

$$\begin{array}{l} v(0)=\\ \exp(0)=\\ 1v'(0)=\\ w(0)\cdot\\ 1=\\ 0\\ v''+\\ Lv=\\ 0v(0)=\\ 1v'(0)=\\ 0L(t)\geq\\ K(t)\\ u''+\\ Ku=\\ 0\\ K\\ p\\ \gamma:\\ [0,l]\rightarrow\\ S\\ S\\ \gamma(l)\\ \gamma(0)\\ w_0\in\\ T_{\gamma(0)}(S)w_1\in\\ T_{\gamma(l)}(S)\\ \gamma\\ J(s)\\ J(0)=\\ w_0J(l)=\\ w_1\\ (J(0),J'(0))\\ J(0)=\\ w_0J(l)=\\ w_1\\ \gamma(l)\\ \gamma(0)\\ (J(0),J'(0))\mapsto\\ (J(0),J(l))\\ \frac{D^2J}{ds^2}+\\ KJ=\\ 0\\ \{e_1(s),e_2(s)\}e_1(0)=\\ \gamma'(0)\\ J(s)=\\ a(s)e_1(s)+\\ b(s)e_2(s)\\ b\\ b''+\\ K(s)b=\\ 0\\ a''+\\ K(s)a=\\ 0\\ a(0),a(l),b(0),b(l)\\ w_0\equiv\\ w_1=\\ 0\\ \gamma(l)\\ \gamma(0)\\ J(s)\\ \gamma:\\ [0,l]\rightarrow\\ S\\ \langle J(0),\gamma'(0)\rangle=\\ 0\\ J'(0)=\\ 0\\ \langle J(s),\gamma'(s)\rangle=\\ 0\\ J(0)\\ \gamma'(0)\\ J'(0)\\ \gamma'(0) \end{array}$$

$$\frac{\eta_b}{2\pi}$$

$$w=\frac{\varphi(l)-\varphi(0)}{2\pi}=\frac{1}{2\pi}\int_0^l(a\,b'-b\,a')\,dt.$$

$$\begin{array}{l} \alpha\colon[0,l]\rightarrow\\ R^2\\ \beta\colon[0,l]\rightarrow\\ R^2\\ p_0\in\\ R^2\\ p_0\not\in\\ \alpha([0,l])\\ p_0\not\in\\ \beta([0,l])\\ t\in\\ [0,l]\\ \alpha(t)\\ \beta(t)\\ \alpha(t)\\ p_0 \end{array}$$

$$|\alpha(t)-\beta(t)|<|\alpha(t)-p_0|.$$

$$\begin{array}{l} \alpha\\ p_0\\ \beta\\ p_0\\ p_0\\ p_0\\ w=\frac{1}{2\pi}\int(a\,b'-\\ b\,a')\,dt\\ \alpha(t)\\ \beta(t)\\ |\alpha(t)-\\ \beta(t)|<\\ |\alpha(t)-\\ p_0|\\ u_\alpha(t)=\\ \frac{\alpha(t)-p_0}{|\alpha(t)-p_0|}\\ u_\beta(t)=\\ \frac{\beta(t)-p_0}{|\beta(t)-p_0|}\\ w(\alpha)=\\ w(\beta)\\ C\\ C\\ K\subset\\ R^2\\ K\\ p,q\in\\ K\\ pq\\ K\\ C\\ C\\ C\\ C\\ K\\ p,q\in\\ K\\ L\\ pq\\ LK\\ L\\ C\\ p\\ q\\ C\\ L\subseteq\\ K\\ K\\ C\\ D\\ C\\ a,b\in\\ C\\ abD\\ c\in\\ ab\\ C\\ C\\ C\\ C\\ K\\ p_0\in\\ K\\ p_0\not\in\\ C\\ p_0\in\\ q\in \end{array}$$